Dr. Juliane Hennecke

Chair of Empirical Microeconomics

MARTIN-LUTHER-UNIVERSITÄT
HALLE-WITTENBERG

ADVANCED MICROECONOMICS

General Equilibrium I - Exchange Economy

Winter Term 2025/2026

Outline

Literature

- Autor, David (2016): Lecture Notes 10
- Varian (2020): Chapter 31 (Exchange)

Overview

General Equilibrium

- So far, we have focused on individual demand for specific products and a **single market at a time** (partial equilibrium)
- In reality, **markets are inter-related**, i.e., changes in one market directly impact other markets.
- **Stylized Example: Bad Coffee Harvest $\rightarrow$ Coffee Price $\uparrow$**
    - **Direct effect:** Households reduce coffee consumption.
    - Demand for **substitutes** (tea, soft drinks) increases $\rightarrow$ their prices rise;
    - demand for **complements** (milk, sugar, coffee machines) decreases $\rightarrow$ their prices fall.
    - **Production-side adjustments:**
        - Tea producers expand and demand more labor.
        - Coffee sector decreases labor demand.
    - **Feedback loops:** Changing wages and prices shift budgets $\rightarrow$ households re-optimize their *entire* consumption bundle.

General Equilibrium

General equilibrium theory (GET)

GET studies the "whole economy" in a micro-founded approach, based on behavior of individual agents. More specifically, it studies a situation in which all markets are simultaneously in equilibrium.

- GET builds the bridges between microeconomics and macroeconomics
- Model that can accommodate the **interactions of all markets simultaneously** in order to determine the properties of the grand equilibrium
- Theory dates back to 1870s and work of French economist **Leon Walras**.
- Theory is used today in **micro-based macroeconomics**: Computable General Equilibrium (CGE) and Dynamic Stochastic General Equilibrium (DSGE) models.

General Equilibrium

- Related questions:
  - Does a general equilibrium **exist**? *(i.e., is there an allocation for which for all markets in the economy supply = demand?)*
  - Under which **conditions** does the general equilibrium exist?
  - Which **properties** does the general equilibrium fulfill?
  - How does the economy **reach** this general equilibrium?
  - Is the equilibrium **fair**?
  - Is the equilibrium **realistic**?

Dimensions of GET

- GET not tractable without complicated computational analysis $\rightarrow$ reduce dimensions of the model without sacrifices of the essence of the problem

$\rightarrow$ Economy with only 2 dimensions:

1. Exchange Economy: Analysis of an economy without production: 2 goods and 2 consumers
2. Robinson Crusoe Economy: Analysis with only one person (Robinson) who produces and supplies his own labor
    - Will not be discussed in this lecture (as focus is on consumption markets) *(If interested: Varian (2022) - Chapter 32)*

Overview

Exchange Economy

- Simple illustration of general equilibrium by focusing on
  - an **exchange economy** without production
  - **two goods** (food (F) and shelter (S))
  - **two individuals** (A and B)
- goes back to Irish economist Francis Ysidro Edgeworth (1845–1926)

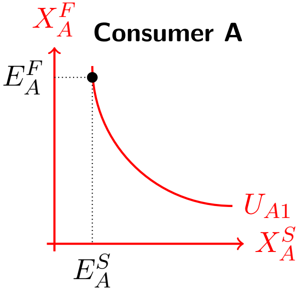
Edgeworth Box

Visual demonstration of a market with just two commodities, X and Y, and two consumers (exchange economy)

- Depiction as a box developed by **Pareto** *(original depiction of Edgeworth is a two-axis graph)*; named after Edgeworth by Marshall

Separate Consumer Problems

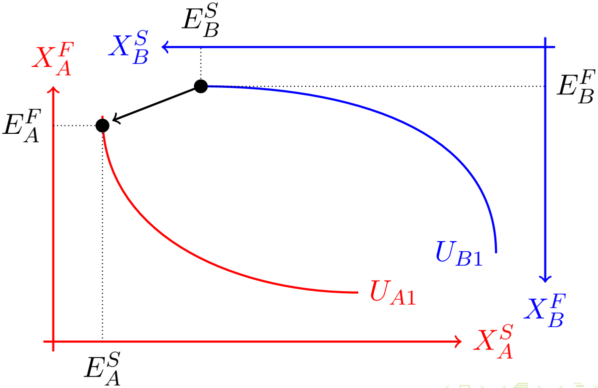
- Economy with two consumers (A, B) and two goods food and shelter (F, S)
- Indifference curves (as we know them) can be used to represent preferences



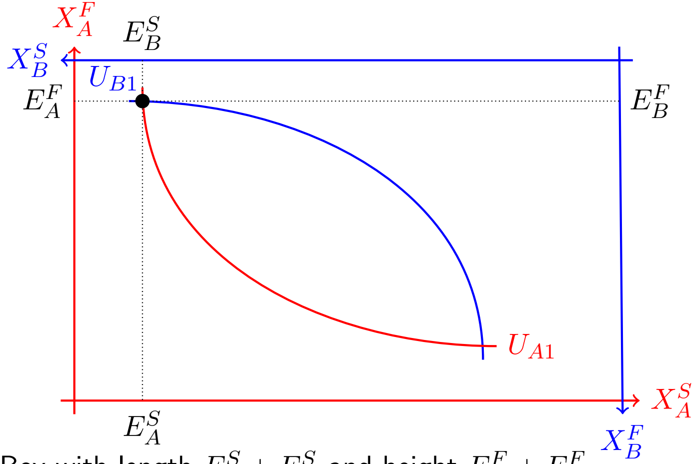
- X_j^i is consumer j's **consumption** of good i
- E_j^i is **initial endowment** of $j \in (A, B)$ with good $i \in (F, S)$

Transforming Separate Problems into Edgeworth Box

- **Exchange economy** with two consumers (A, B) and two goods (F, S) and four endowment $(E_A^F, E_A^S, E_B^F, E_B^S)$
  - Indifference curve from consumer B is turned upside down
  - Consumption of consumer A seen from left bottom edges of box, consumption of consumer B from right top edges.



Edgeworth Box



- Box with length $E_A^S + E_B^S$ and height $E_A^F + E_B^F$

Set-Up of the Edgeworth Box

- An **allocation** is a consumption pair

$$\left(X_A = (X_A^F, X_A^S), X_B = (X_B^F, X_B^S)\right)$$

- Without trade, consumption bundles equal endowment

$$X_A = E_A$$
$$X_B = E_B$$

- *With trade*, total consumption of each good has to equal total endowment of each good:
  - An allocation is **feasible**, if

$$X_A^F + X_B^F = E_A^F + E_B^F$$
$$X_A^S + X_B^S = E_S^A + E_B^S$$

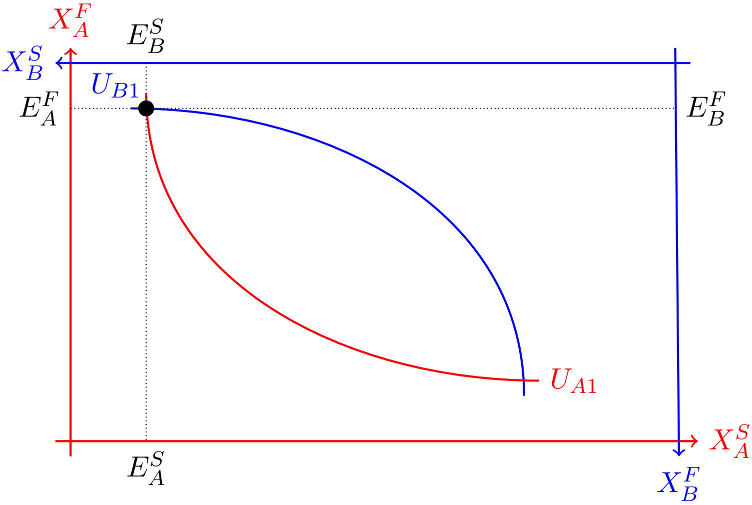
  - All feasible allocations are included in this box.

Overview

Assumptions: Market Conditions

- Any trade taking place satisfies four assumptions about market conditions:
  1. **No transaction costs**: No F or S is destroyed by exchanging/trading F or S.
  2. **No market power**: A and B are **price takers-**
  3. **No externalities**: A's utility only depends on X_A^F, X_A^S, not on consumption of B and vice versa.
  4. **Full and symmetric information**: A and B have full information, e.g. on quality of goods.
- We also call these conditions of **perfect market**.

Trades in the Edgeworth Box



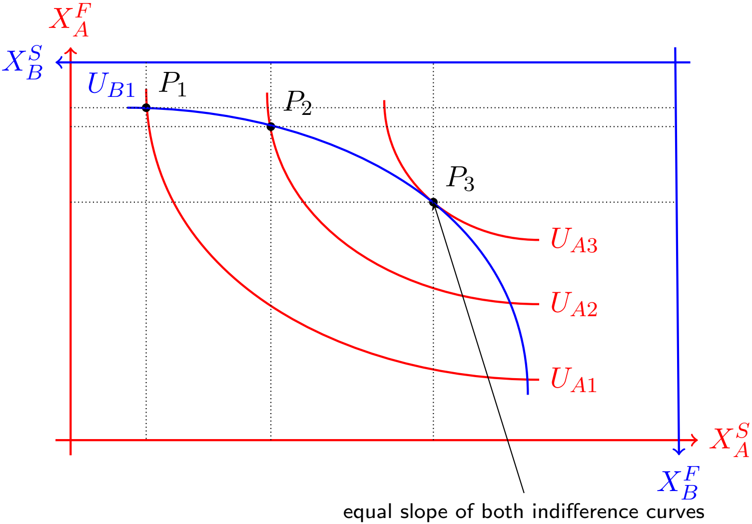
Will A and B trade?

- All allocations in the lens between indifference curves of A and B through their initial endowment are preferred by A and B compared to initial endowment.
- The allocations in this lens are a **Pareto improvement** over the initial endowment, i.e., they increase utility of A and/or B without making anyone worse off as they **Pareto dominate** the starting point.

$\rightarrow$ **Gains from trade**

- If there are gains from trade, consumers will engage in trade up to the point where no more Pareto improvements are possible.

Exchange optimum that leaves A as well of as possible



Trades between A and B

- Starting from initial endowment (P_1) A has a lot of food (F) but not a lot of space to live (S).
- A would be better off when exchanging food for shelter.
- E.g. by going to point P_2.
- But even at P_2, gains from trade remain.
- All gains from trade are exhausted when the two indifferent curves are tangent to each other.

Overview

Pareto Efficiency

Pareto efficient allocation

Allocation in which it is not possible to make one person better off without making at least one other person worse off

- At the Pareto efficient allocation, all gains from trade are exhausted.
- In the Edgeworth Box all Pareto efficient allocations are allocations at which the indifference curves of the two consumers are **tangent** to each other.
- No one of the two consumers can reach a higher indifference curve, without putting the other consumer on a lower one.

Pareto efficient allocation

- At a Pareto efficient allocation, the **indifference curves have to be tangent**:

$$MRS^A = -\frac{\frac{\partial u^A}{\partial X^F}}{\frac{\partial u^A}{\partial X^S}} = -\frac{\frac{\partial u^B}{\partial X^F}}{\frac{\partial u^B}{\partial X^S}} = MRS^B$$

- Assume this condition wouldn't hold, e.g. $MRS^A = -1$ and $MRS^B = -2$
  - Consumer A would be willing to exchange 1 unit of F against 1 unit of S, while B would be willing to give up 2 units of S for one unit of F.
  - There'd be gains from trade as A would be better off if she got two units of S and gave B one unit of F, while B would not be harmed.
- An allocation with $MRS^A \neq MRS^B$ is not Pareto efficient.

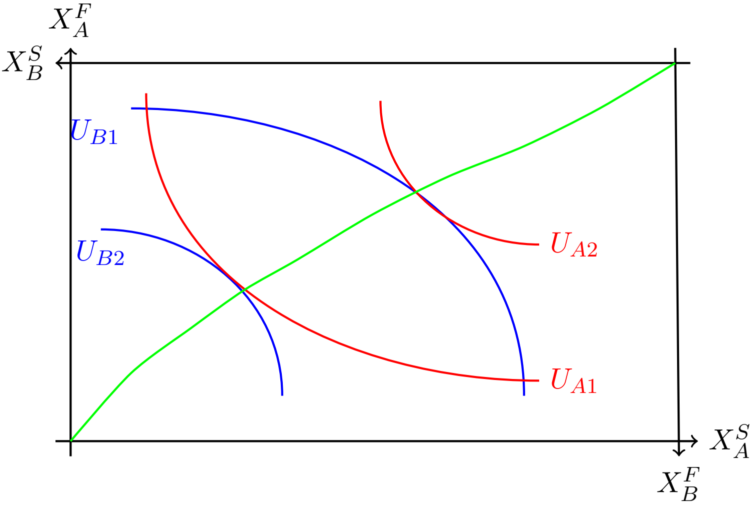
Contract Curve

- Generally, there are a lot of Pareto efficient allocations.
- If both consumers have strictly monotone preferences then the set of all Pareto efficient allocation is a curve from bottom left to top right of the Edgeworth Box.

Contract Curve / Trade Curve

Set of all Pareto efficient allocation (pareto set)

Contract Curve



Overview

How does trade work in the Edgeworth Box?

- Which point **on the contract curve** will be reached through trade?
  - Only points on the contract curve *within* the lens between the original indifference curves can be reached with pareto improvements from the **endowment point**
- But which point **on the contract curve within the lens** will be reached through trade?
  - To get at this, we can impose **rules of trade**.
  - We now introduce **prices** in this economy that are set so that they **clear the market**.
  - We assume (as stated above) that the two consumers act as price takers.

The Walrasian Auctioneer

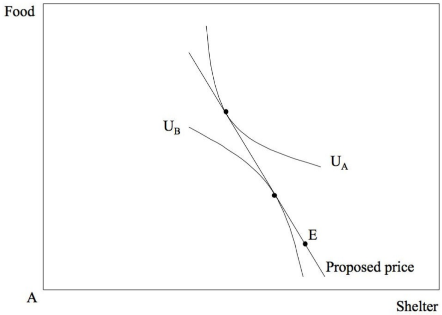
Walrasian Auctioneer

Auctioneer who sets the prices in the economy which clear the market, i.e. acts as an intermediary to find a market-clearing price.

1. At the initial endowment, the market clears (all goods are consumed) but the allocation is not Pareto efficient.
2. The auctioneer now sets prices for F (P_F) and S (P_S).
3. Both consumers take these prices as given and announce how much of F and S they'd want to consume at these prices.

The Walrasian Auctioneer

4. But of course, the auctioneer does not necessarily know which prices will clear the market.



Source: Autor (2016), LN 10, p. 7

- For the price vector called by the auctioneer, A wants less S and more F and B more S and less F.
- But changes wanted by A are larger than the changes wanted by B:
- There is **excess demand** for F $(X_A^F + X_B^F > E_A^F + E_B^F)$ and **excess supply** of S $(X_A^S + X_B^S < E_A^S + E_B^S)$.

$\rightarrow$ Market is in **disequilibrium**

The Walrasian Auctioneer

5. If the markets don't clear, the auctioneer will increase price of good with excess demand and decrease price of good with excess supply.
6. ...
7. Trade happens when auctioneer has found price vector (P_F, P_S) for which both markets (for F and S) clear.

Is there a Walrasian Auctioneer in reality?

- Answer: (Of course) no.
- But there are good news: the auctioneer is not needed to reach the equilibrium.
- One can show that the equilibrium can also be reached *by the market itself* through a **process of "Tâtonnement"** (slowly-approaching, groping).

Walras Equilibrium

- **Walras Equilibrium:** The exchange mechanisms results in a **Pareto efficient equilibrium** (an allocation of consumption bundles and prices for a specific initial allocation)
- Total Demand equals total supply:

$$
\begin{aligned}
X_A^F(p_F^*, p_S^*) + X_B^F(p_F^*, p_S^*) &= E_A^F + E_B^F \\
X_A^S(p_F^*, p_S^*) + X_B^S(p_F^*, p_S^*) &= E_A^S + E_B^S
\end{aligned}
$$

$\rightarrow$ Allocation is feasible - no excess demand

- **Aggregated excess demand** of both F and S is zero

$$
\begin{aligned}
Z_F(p_F^*, p_S^*) = X_A^F(p_F^*, p_S^*) + X_B^F(p_F^*, p_S^*) - E_A^F - E_B^F &= 0 \\
Z_S(p_F^*, p_S^*) = X_A^S(p_F^*, p_S^*) + X_B^S(p_F^*, p_S^*) - E_A^S - E_B^S &= 0
\end{aligned}
$$

Walras Law

- If the condition $Z(p_F^*, p_S^*) = 0$ (zero excess demand) holds for one good, it automatically holds for the other one, which follows from Walras Law

Walras Law

The value of the aggregated excess demand is identical to zero.

$$p_F Z_F(p_F^*, p_S^*) + p_S Z_S(p_F^*, p_S^*) = 0$$

- Walras Law holds for **all possible choices of prices** (not just equilibrium prices)
  - Proof follows from adding up the budget constraints of both consumers *(see Varian (2021), p.574)*

Existence of the Equilibrium?

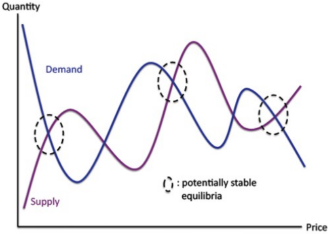
- Why is it **always Pareto efficient**?
  - A and B optimize for the same price vector (P_F/P_S), i.e., for A and B holds slope of indifference curve $= P_F/P_S$.
  - Further, as the price ratio separates the choice sets of A and B, the indifference curves of A and B are tangent (and there are no further gains from trade).
- Does this price vector (that leads to a Pareto-efficient allocation) exist?
  - Answer: Yes, as long as consumers' preferences are convex, i.e., we have a decreasing marginal rate of substitution.
  - Thus the Walrasian Equilibrium on markets for F and S exists and is Pareto efficient if
    - market conditions hold.
    - consumers have convex preferences.

Uniqueness of the Equilibrium?

- Is this Walrasian equilibrium unique? - **No!**

Sonnenschein-Mantel-Debreu Theorem

The excess demand function can take any shape and the equilibrium is therefore not unique (unless additional assumptions are made).



- **"Law of Demand"** ($\frac{\partial x}{\partial p_x} < 0$) applies to individual Hicksian demand curves
- This does not apply to **market demand curves**

Overview

Summary

- General equilibrium analyzes how all markets interact simultaneously – not one market in isolation.
- An exchange economy can be fully described by preferences and endowments of consumers.
- The Edgeworth Box is a powerful tool to visualize feasible allocations, indifference curves, and efficiency.
- Pareto-efficiency requires equality of marginal rates of substitution ($\mathrm{MRS}_\mathrm{A} = \mathrm{MRS}_\mathrm{B}$).
- The contract curve represents all Pareto-efficient allocations in the economy.
- Market-clearing prices ensure that supply equals demand for all goods simultaneously, no excess demand.
- A Walrasian (competitive) equilibrium is a set of prices and allocations where all markets clear and all consumers optimize.

Important Keywords

1. Partial vs. General Equilibrium
2. Exchange Economy
3. Endowments
4. Edgeworth Box
5. Pareto Improvements, Pareto Efficiency
6. Contract Curve
7. Excess Demand Curve, Market Clearing
8. Walrasian Auctioneer
9. Walrasian (General) Equilibrium
10. Conditions of Perfect Markets

Questions?